8 Fig. 8.1 shows images produced during two different medical scanning procedures.



ultrasound scan of a fetus



X-ray scan of a hand

Fig. 8.1

(a) (i) Define ultrasound.

.....

..... [1]

(ii) State how the speed of sound in liquid compares to the speed of sound in air.

..... [1]

(iii) X-rays are part of the electromagnetic spectrum.

State the speed of X-rays in a vacuum.

..... [1]

(b) Describe **three** similarities or differences between the use of ultrasound and X-rays in medical scanning procedures.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 6]